

Article

An Adaptive Optical Limiter Based on a VO₂/GaN Thin Film for Infrared Lasers

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Abstract

Vanadium dioxide (VO₂) is a highly promising material for infrared laser protection due to the pronounced optical switching effect during its metal–insulator transition (MIT). However, due to the relatively high MIT temperature of VO₂ and the low transmittance contrast before and after the MIT, practical applications face challenges in modulation depth and response time. In this study, we address these issues using a wafer-scale VO₂/GaN/Al₂O₃ heterostructure fabricated by oxide molecular beam epitaxy. The conductive GaN interlayer enables local Joule heating of the VO₂ film, permitting direct control of the MIT via an external bias with a threshold of 4.7 V. This structure exhibits a substantial resistance change of four orders of magnitude and enables adaptive limiting of a 3.7 μm laser, reducing transmittance from 60% to 10%. Our work demonstrates a practical, wafer-scale laser-protection device and introduces a pre-excitation strategy via external biasing to enhance response performance.

Keywords: VO₂; infrared laser protection; metal-to-insulator transition

1. Introduction

As a strongly correlated electron oxide, VO₂ undergoes a distinct metal–insulator transition (MIT) upon temperature variation, transitioning from an insulating or semiconducting phase at lower temperatures to a metallic phase at elevated temperatures [1]. In addition, a distinct optical switching effect, especially within the infrared wavelength range, always appears across the MIT process [2,3]. Due to these special electronic and optical properties, VO₂ demonstrates tremendous potential applications in many fields, such as smart windows [4], optical switching devices [5], memory materials [6], photoconductive infrared detectors [7], and thermal [8]/chemical sensors [9,10].

The pronounced switching behavior of VO₂ within the infrared spectral region renders it highly suitable for application as an infrared photoelectric limiter. The most important performance parameter for the infrared limiter is the switching characteristic of its infrared transmittance. Therefore, many methods have recently been introduced to improve the



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infrared photoelectric switching properties of VO₂ films [11–17]. Wang et al. [18] developed a tunable infrared optical switch by integrating a VO₂ film within an Al nanoarray plasmonic structure, achieving a modulation depth as high as 99.4% and an extinction ratio exceeding −22.16 dB. Wan et al. [19] designed a VO₂ inductive transfer filter (ITF) that modulates the response from single-band to multi-band and improves infrared imaging contrast. Besides thermal stimulation, alternative stimuli including electric fields (bias voltage or current) and electromagnetic radiation are also effective in triggering the MIT in VO₂ to meet the necessary conditions for its practical application and provide more potential applications [20–23]. It is noteworthy that for high-speed, linear modulation applications, transparent conductive oxides such as indium tin oxide (ITO) have demonstrated remarkable potential, achieving GHz-rate operation through carrier-plasma dispersion effects [24]. For instance, Qi et al. [25] prepared a VO₂/ITO limiter, with a threshold voltage of only 1.4 V, a modulation depth of 35% of the optical switch, and response times of 0.24 ms and 2.1 ms, respectively. Optical limiters, as a class of nonlinear photonic devices, are characterized by a reduction in optical transmittance as the intensity of the incident light increases [26]. Within a silicon photonics platform, Parra et al. [27] implemented an integrated optical limiter employing a miniaturized VO₂/Si waveguide configuration that exhibits an activation threshold as minimal as 3.5 mW. It is reported that the phase transition of VO₂ can be induced by lasers [28,29], which enables the limiting of infrared lasers. Employing micro-Raman spectroscopy with gradually increasing excitation power, Vilanova-Martínez et al. [30] studied the impact of intense visible laser radiation on VO₂ crystals maintained in an ambient air environment, and achieved a laser-induced reversible phase transition of VO₂. However, despite the progress achieved in VO₂ films in terms of composite films, element doping and other aspects, the synergistic optimization of a reduced phase transition threshold and a high infrared switching ratio remains a significant challenge for VO₂ films.

In this work, a VO₂/GaN/Al₂O₃ composite film was fabricated via metal–organic chemical vapor deposition (MOCVD) and molecular beam epitaxy (MBE). A pre-excitation strategy with localized temperature control via an external voltage was proposed. It successfully reduces the threshold power to 330 mW and shortens the response time to 9 s, effectively breaking through the performance bottlenecks of traditional schemes. The conductive GaN interlayer enables local Joule heating of the VO₂ film, permitting direct control of the MIT via an external bias with a threshold of 4.7 V. This structure exhibits a substantial resistance change of four orders of magnitude and enables adaptive limiting of a 3.7 μm laser, reducing transmittance from 60% to 10%.

2. Materials and Methods

The core of this device for rapid adaptive optical limiting lies in the adoption of a pre-excitation strategy, and its working principle is illustrated in Figure S5. Prior to laser incidence, a preset fixed DC voltage is first applied to the intermediate GaN layer. At this point, the conductive GaN layer generates Joule heat, heating the VO₂ film to a pre-activated state close to the phase transition temperature. In this state, when irradiated by a mid-infrared laser, the film only needs to absorb a very small amount of additional light energy for its temperature to quickly cross the phase transition point, triggering a rapid transition from the high-transmission insulating state to the low-transmission metallic state, thereby achieving effective laser limiting. Once the laser irradiation ceases, the applied voltage is removed, and the device cools naturally through heat conduction, ultimately returning to the initial high-transmission insulating state. This completes an entire working cycle, and the device is ready for the next round of protection.

A commercial conductive n-type GaN epitaxial layer on a c-plane sapphire grown by MOCVD was used in this work. The GaN substrates were ultrasonically cleaned in acetone and isopropanol sequentially for 10 min each, which was followed by a deionized water rinse. Then, the VO₂ layer was grown on this n-GaN layer by an rf-plasma-assisted oxide-MBE. And 4 N-purity vanadium evaporated by an e-beam evaporator and 5 N-purity O₂ activated by a radio frequency plasma source were used for oxygen flux. Throughout the epitaxial synthesis of the VO₂ layer, the substrate temperature was stabilized at 530 °C and the deposition chamber pressure sustained at 3.2×10^{-5} torr. Concurrently, reflection high-energy electron diffraction (RHEED) was utilized to monitor the structural evolution in situ during the entire growth process.

The microstructural properties of the synthesized epitaxial thin films were probed via Raman spectroscopy employing a 532 nm excitation source. Furthermore, crystallographic phases and epitaxial alignment were determined through X-ray diffraction (XRD) characterization conducted at the BL14B1 beamline of the Shanghai Synchrotron Radiation Facility (SSRF). ϕ scan XRD was also performed in order to examine the epitaxial growth behavior at the interface. The resistance of the film sample, as a function of temperature, was characterized by employing a customized four-point probe configuration integrated with a precision temperature-regulated sample platform.

3. Results and Discussion

The conventional direct growth of VO₂ on thick Al₂O₃ substrates poses a challenge for laser-induced adaptive limiting, as the substrate absorbs most incident energy, preventing the VO₂ from reaching its phase transition temperature. Our solution, depicted in Figure S1, is a multilayer structure incorporating a conductive GaN interlayer. This design serves dual purposes: it ensures epitaxial compatibility for VO₂ growth and enables direct Joule heating of the VO₂ via an applied bias to the GaN, decoupling the heating from laser absorption. To finalize the design, the VO₂ film thickness was optimized through optical simulations. The relationship between transmittance, VO₂ film thickness, and wavelength is visualized in the 3D plot in Figure 1, where the transmittance characteristics of VO₂ in the low-temperature insulating state and high-temperature metallic state are presented in Figure 1a,b, respectively. It can be seen that no drastic abrupt change in transmittance occurs when VO₂ is in the low-temperature insulating state. This phenomenon is consistent with the optical absorption law of semiconductor materials. In the high-temperature metallic state, the transmittance decreases significantly. Particularly in the 2–4 μm band, the transmittance drops rapidly to below 20% when the film thickness exceeds 50 nm. Furthermore, the modulation depth in the mid-infrared region is much larger than that in the near-infrared region. These results not only verify the feasibility of VO₂ as an infrared optical limiter but also provide key references for the selection of the device's operating band and the optimization of film thickness in subsequent work. The measured transmittance spectra for this optimized device before and after phase transition are provided in Figure S2. A VO₂ thin film with a flat and dense surface was prepared on a GaN/Al₂O₃ substrate by MBE, and the details of the epitaxial film preparation have been systematically elaborated on in previously published work [29]. As shown in Figure 2, the surface morphology of the VO₂ film was tested with a scanning electron microscope (SEM), from which it can be seen that the VO₂ film has flat surfaces, dense and uniform film layers, and obvious and tightly bonded particles. From the energy-dispersive spectrometer (EDS) spectrum (Figure S3), it can be seen that the VO₂ film does not contain other elemental impurities, except VO₂ and GaN.

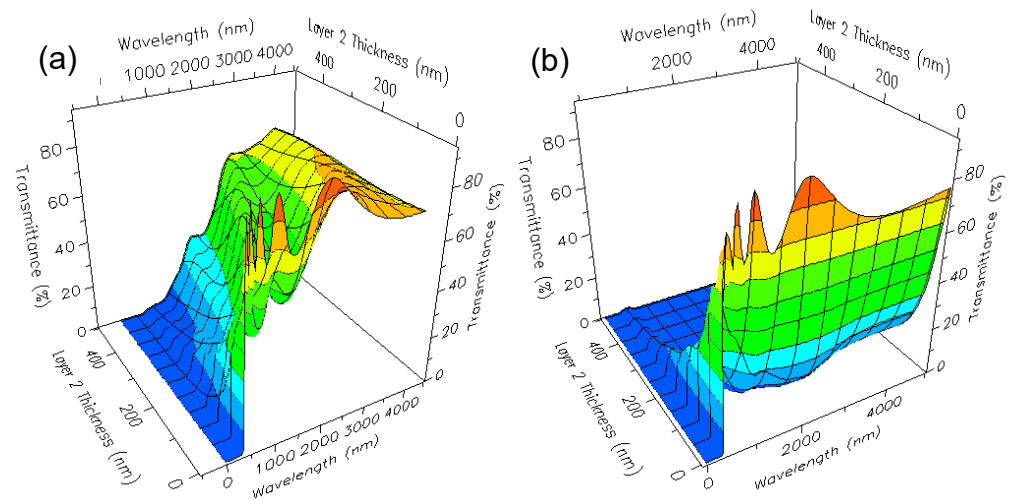


Figure 1. Three-dimensional plot of transmittance versus film thickness. (a) Before VO₂ phase transition; (b) after VO₂ phase transition.

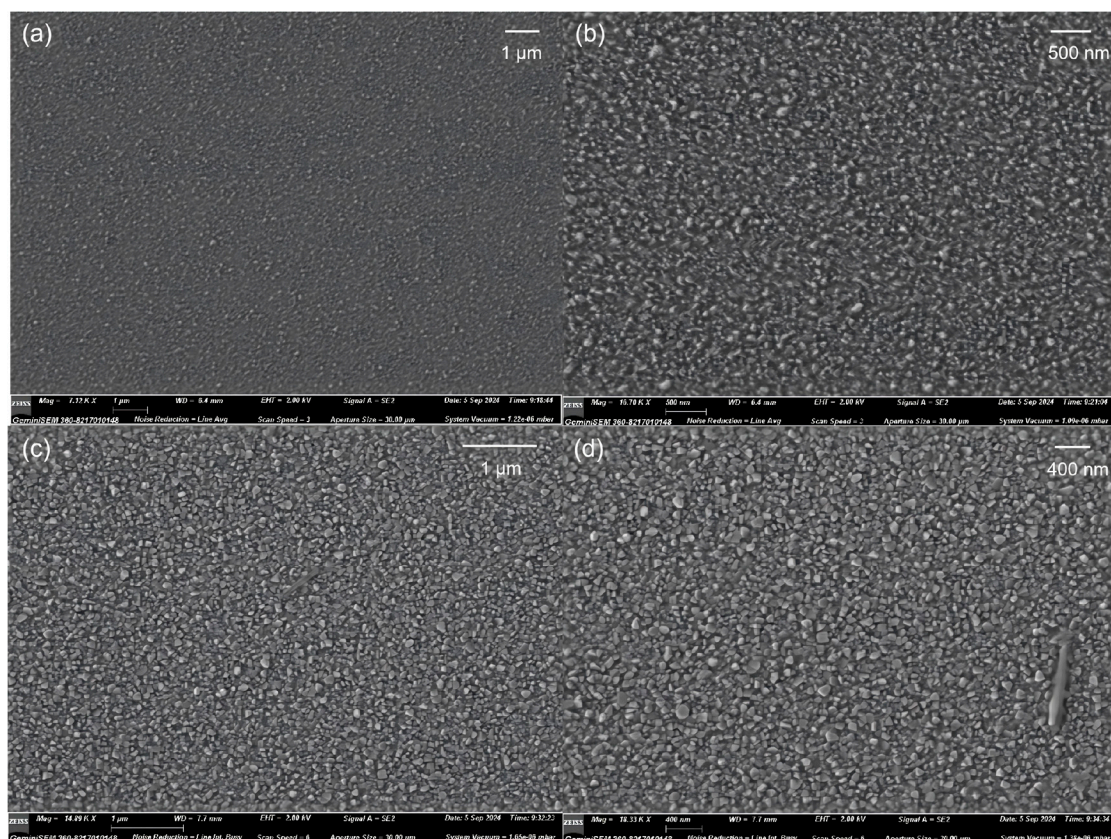


Figure 2. (a–d) SEM image of VO₂ thin film.

In order to study the crystal structure and epitaxial alignment relationships within the VO₂/Ga₂N heterostructure, detailed θ -2 θ scanning and φ scanning diffraction tests were carried out. In the θ -2 θ scanning mode (shown in Figure 3a), it can be seen that the sample exhibits a strong Al₂O₃ single crystal substrate diffraction peak, and the micron-thick Ga₂N film also has a strong Ga₂N (002) diffraction peak. The epitaxial VO₂ film grown on the Ga₂N/Al₂O₃ substrate has a single diffraction peak at $2\theta = 39.8^\circ$, which is the diffraction peak of VO₂ (020) after comparison. And in the φ scanning diffraction studies, considering the hexagonal symmetric structure of the Al₂O₃ (0001) plane, the diffraction signals of the VO₂ (011) and Ga₂N/Al₂O₃ (102) planes were selected in the φ scan mode. It can be seen

from Figure 3b that along the (001) direction, the Al_2O_3 single crystal has triple symmetry, while the GaN film has a sixfold symmetric structure. In addition, it can be seen that there is an in-plane rotation angle of 30° between the GaN (102) diffraction peak and the Al_2O_3 (102) in the φ scan mode. By comparing the diffraction peaks of VO_2 (011), we can see the hexagonal symmetry, and the diffraction peaks of VO_2 (011) correspond to the diffraction peaks of GaN (102), indicating that the a-axis direction of VO_2 is consistent with the b-axis direction of GaN in the plane. Thus, the lattice orientation match in the plane can be written as Al_2O_3 ($-12-10$)/GaN (0-110), Al_2O_3 (10-10)/GaN (2-1-10) or Al_2O_3 (11-20)/GaN (0-110), Al_2O_3 (1-100)/GaN (2-1-10) and VO_2 (100)/GaN ($-12-10$), VO_2 (001)/GaN (2-1-10). The in-plane matching relationship of this epitaxial growth is consistent with that reported in the literature, indicating that this sample has very good epitaxial growth behavior.

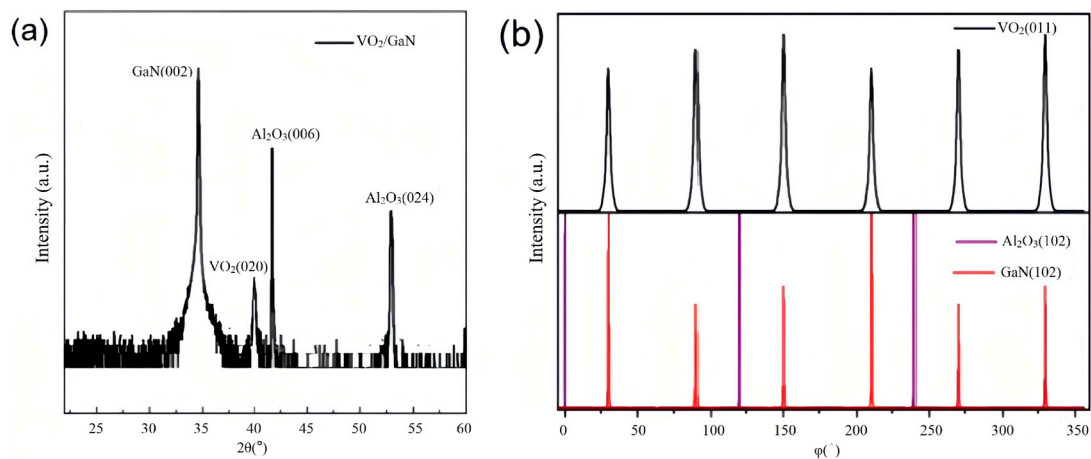


Figure 3. (a) XRD pattern of VO_2/GaN sample in θ - 2θ scan mode; (b) φ scan mode of sample.

Temperature-variable Raman spectroscopy can directly detect the structural transition of VO_2 films from the low-temperature monoclinic phase to the high-temperature rutile phase before and after the MIT. Figure 4a shows the Raman spectrum of the sample during the heating process. It can be seen that at 30°C , except for the Raman peak from the substrate, the VO_2 in the low-temperature insulation phase presents the characteristic Raman peaks of a monoclinic phase structure at 196 cm^{-1} , 225 cm^{-1} , 312 cm^{-1} , 395 cm^{-1} and 617 cm^{-1} . As the temperature rises, the characteristic Raman peaks of VO_2 gradually weaken, and they disappear completely at a temperature of 70°C or higher, which indicates that VO_2 has undergone a complete phase transition and the entire crystal structure has completely changed to the rutile metallic phase. Figure 4b shows the cooling process starting from 90°C . It can be seen that with the decrease in temperature, the characteristic Raman peak of the low-temperature insulating phase of VO_2 gradually begins to appear at 60°C or lower temperatures. The results show that the low-temperature monoclinic phase begins to appear again during the cooling process, which proves the reversibility of the temperature-modulated metal-insulation phase transition of VO_2 . At the same time, it can be seen from the Raman spectrum of the variable temperature that the phase transition temperature corresponding to the heating and cooling process is between 60°C and 70°C , which is in good agreement with the critical temperature obtained by the variable temperature resistance test.

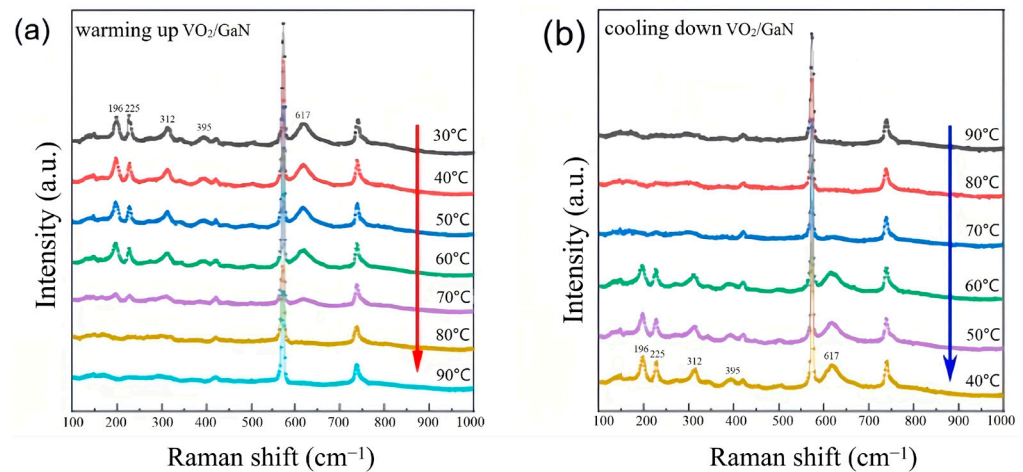


Figure 4. Raman spectrum of VO₂/GaN sample (a) under heating; (b) under cooling.

In order to further investigate the influence of VO₂ film phase transition on its photoelectric properties, we conducted a temperature variation performance test on the device (as shown in Figure 5). It can be seen from Figure 5 that the sample has a large resistance ($\sim 6 \times 10^5 \Omega$) at room temperature, showing an insulating state. As the temperature increases, the resistance changes by nearly four orders of magnitude, thus showing a metallic state ($\sim 80 \Omega$). During the cooling process, the sample reverts from the high-temperature metallic state to the room-temperature insulating state. The temperature hysteresis effect is obvious in the process of increasing and decreasing temperature, which is a typical phase transition characteristic of vanadium dioxide phase change materials. According to the differential graphs of the heating and cooling curves shown in the inserted figure, the critical phase transition temperatures corresponding to the heating and cooling processes of the prepared VO₂/GaN film are 68 °C and 61 °C, respectively, which are in good agreement with the previous literature reports.

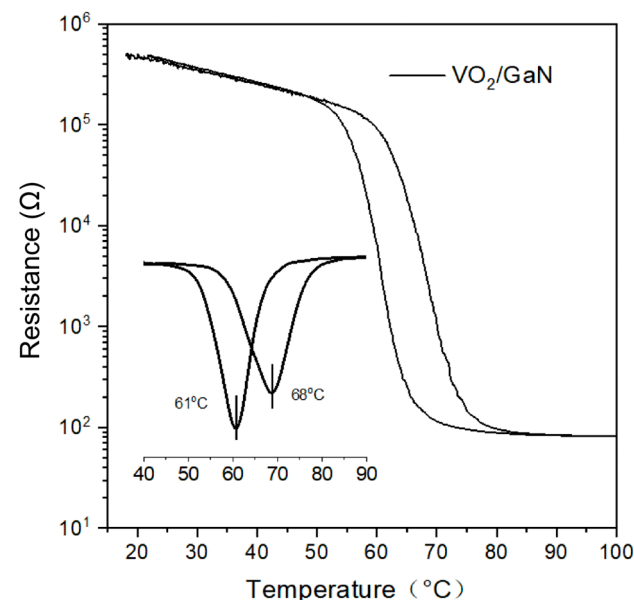


Figure 5. Temperature-variable resistance test of the VO₂/GaN film samples; the lower left insert is the differential graph of the heating and cooling curves.

The relationship between the device's mid-infrared transmittance and the applied bias voltage was characterized using a 3.7 μm laser (setup shown in Figure S4). As shown in Figure 6, the transmittance initially decreased gradually with increasing voltage until

reaching approximately 4.7 V, where a sharp drop occurred, indicating a phase transition in the VO₂ film. This transition was marked by a substantial increase in reflectance and a concurrent drastic decrease in transmittance, establishing ~4.7 V as the critical phase transition voltage during heating. Conversely, during the downward voltage sweep, the transmittance recovered abruptly at around 3.6 V. To further verify the localized Joule heating effect of the GaN interlayer, we have added supplementary infrared thermal images, as shown in Figure S5.

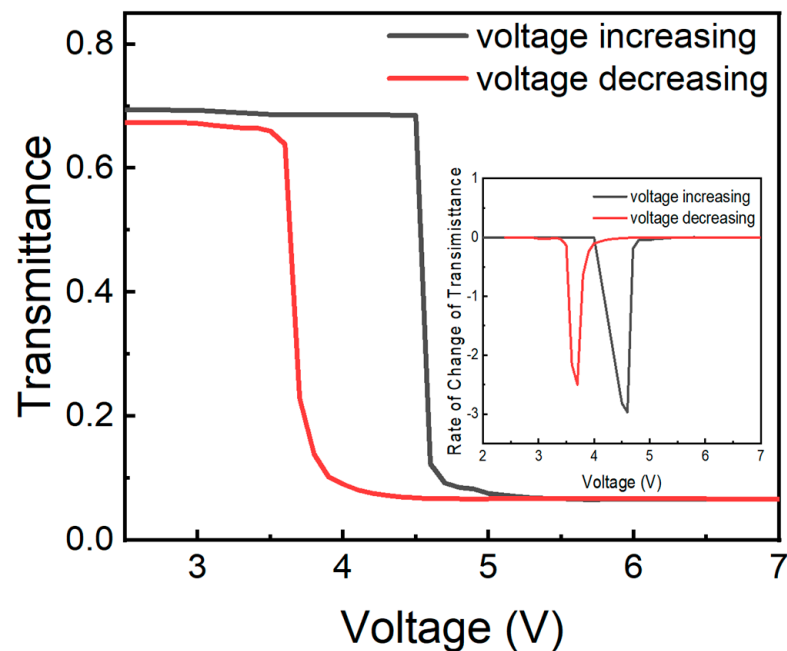


Figure 6. Transmittance as a function of the electric field applied to GaN.

To evaluate the device's adaptive amplitude-limiting response to mid-infrared lasers, we employed the same optical configuration (Figure S2). As shown in Figure 7a, the output power of the VO₂/GaN/Al₂O₃ composite film is plotted against the incident laser power for various bias voltages. The laser power required to trigger the phase transition decreases with increasing bias. For example, the output power begins to saturate at ~780 mW incident power under 2.9 V bias, while under 4.0 V bias, the transition occurs at a significantly lower incident power of ~260 mW. Figure 7b presents the mid-infrared transmittance as a function of incident laser power under the same conditions, directly showing the sharp transmittance drop from ~50% to ~10% upon phase transition and its shift to lower incident powers at higher bias. The combination of these two results clearly demonstrates that electrical pre-excitation via the GaN layer significantly enhances the sensitivity of the optical limiter.

Cycle stability is a critical performance metric for laser limiting devices. Figure 8a presents the film transmittance as a function of the square-wave pulse voltage period, revealing clear repeatability and stability during the phase transition process. Figure 8b shows the transmittance variation over a single voltage pulse cycle. Upon phase transition, the transmittance drops from 60% to 10%, with measured response times of 9 s for the "on" state and 37 s for the "off" state. Although the transmittance decreases shortly after the voltage is applied, the optical switch remains in the "off" state for a certain duration even after the electrical excitation is removed. This behavior can be attributed to the thermal hysteresis effect induced by Joule heating and the heat dissipation limitation of the multilayer heterostructure (as shown in Item 8 of the Supporting Information), suggesting

that thermally driven phase transition plays an important role in large-scale VO₂-based electro-optic switches, ultimately limiting their “off” response speed.

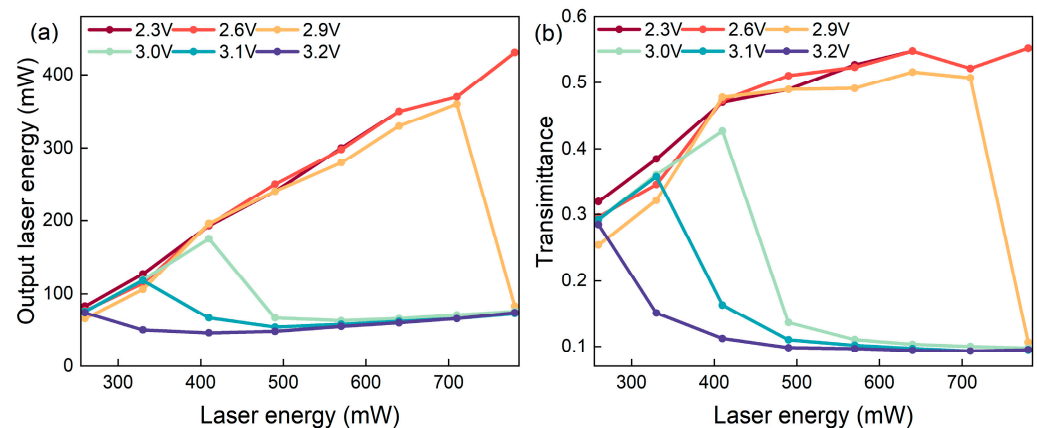


Figure 7. (a) Relationship between the output power and incident laser energy of the composite film under different voltages; (b) relationship between the infrared transmittance and incident laser energy of the composite film under different voltages.

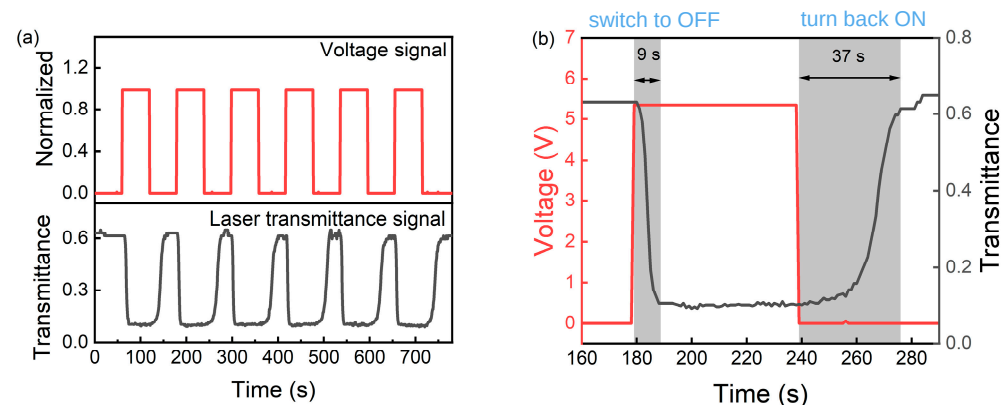


Figure 8. (a) The voltage signal and transmittance variation with time in the VO₂/GaN bilayer; (b) the change in film transmittance for one periodic square-wave voltage pulse.

Laser-induced phase transition capability and cycling durability are essential for VO₂-based adaptive laser limiters. As shown in Figure 9, the device was subjected to periodic laser switching (300 s intervals) under a 3.5 V bias to evaluate its adaptive phase transition behavior. When the laser is turned on, the transmittance decreases from approximately 60% to 10%, and this transition is reproducibly observed over multiple cycles, demonstrating the excellent repeatability and cycle stability required for adaptive laser limiting applications.

As shown in Table 1, we have summarized and classified the performance indicators of some VO₂ laser limiter devices. Compared with other VO₂-based tunable systems, this work has for the first time achieved wafer-scale fabrication, laying the foundation for the large-scale integration and industrialization of the devices. In terms of performance, this heterostructure can achieve a modulation depth of up to 50% (from 60% to 10%) with only 330 mW (3.5 V) of electrical driving power. It also has excellent response characteristics (rise/fall times are 9 s and 37 s respectively). Particularly importantly, by integrating VO₂ with a high-thermal-conductivity and high-stability GaN epitaxial layer and sapphire substrate, this system demonstrates excellent comprehensive engineering applicability, including high reliability, good thermal management capability, and high compatibility with existing semiconductor processes. These features collectively address the core bottlenecks commonly faced by current VO₂-based dynamic photonic devices,

such as high driving power consumption, limited modulation amplitude, and difficulty in large-scale production.

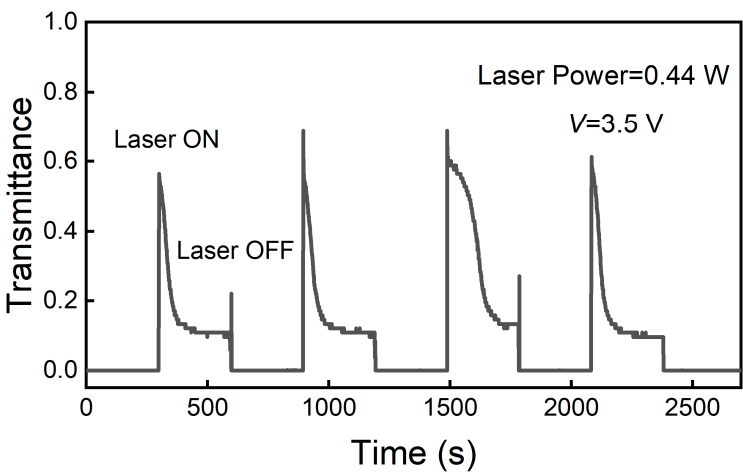


Figure 9. The transmittance changes in the VO₂/GaN device with the laser cycling ON and OFF.

Table 1. Performance comparison between our device and recent electrically controlled VO₂/PCM optical limiters.

Material System	Threshold (Power/Voltage)	Response Time (Rise/Fall)	Modulation Depth	Operating Wavelength	Key Features	Ref.
VO ₂ /ITO film	1.4 V (electrical)	0.24 ms/2.1 ms	35%	Not specified	Low threshold voltage Fast response Small-scale device	[25]
VO ₂ -metal metasurface	Not specified	Not specified	Not specified	Mid-IR	Electrically tunable Nanophotonic structure Limited scalability	[5]
VO ₂ /Si waveguide	3.5 mW (optical)	Not specified	Not specified	Not specified	Ultra-compact Low optical threshold On-chip integration only	[26]
Ultrathin VO ₂ -based limiter	Not specified	Not specified	Broadband	Mid-IR	Broadband reflection Plasma-enhanced Small-area fabrication	[23]
Al nanoarrays/VO ₂	Optical power threshold	Not specified	~99.4%	1.5 μm	Plasmonic resonance-enhanced High modulation depth	[18]
VO ₂ /GaN/Al ₂ O ₃ heterostructure	330 mW (at 3.5 V)	9 s/37 s	50% (60%→10%)	3.7 μm (mid-IR)	Wafer-scale Electrical control High compatibility with practical applications	This work

4. Conclusions

In summary, a VO₂/GaN/Al₂O₃ multilayer composite film was fabricated by growing a VO₂ film on a GaN/Al₂O₃ substrate via molecular beam epitaxy. Four-point probe measurements revealed that the film’s resistance changes by four orders of magnitude between high and low temperatures. Electrically heating the GaN layer enabled the identification of a critical phase transition voltage of 4.7 V for the VO₂ film. Using this, a mid-infrared (3.7 μm) laser adaptive limiter was demonstrated. Upon applying a bias slightly below this threshold and an incident laser energy of 330 mW, the film underwent a phase transition, and its transmittance dropped from 60% to 10%. This result confirms the successful realization of an adaptive infrared laser limiter based on the VO₂/GaN/Al₂O₃ composite film.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/photonics13020148/s1>: Figure S1: Schematic diagram of thin film structure; Figure S2: Comparison of VO₂ film phase transition optical transmittance before and after phase change; Figure S3: EDS spectrum of VO₂ thin film; Figure S4: Optical path of VO₂ film's critical phase transition electric field measurement; Figure S5: Infrared thermal imaging of the VO₂/GaN/Al₂O₃ device under applied voltage; Thermal Analysis and Laser Damage Threshold Evaluation; The Control Logic of the Electrical Pre-Excitation Strategy; Figure S6: Schematic diagram of the working principle for the pre-excitation strategy; Discussion and analysis of the thermal hysteresis and heat dissipation bottlenecks contributing to slow recovery.

Author Contributions: Conceptualization, C.Z. (Changqi Zhou) and J.H.; Methodology, Y.L. and C.Z. (Changqi Zhou); Software, C.Z. (Changqi Zhou); Validation, Y.L., C.Z. (Changqi Zhou), J.Z., S.Z. and C.Z. (Congwen Zou); Formal analysis, S.W. and Y.S.; Investigation, Y.L., S.W., S.Z., J.H. and Y.S.; Resources, C.Z. (Changqi Zhou), W.J. and C.Z. (Congwen Zou); Writing—original draft, Y.L. and C.Z. (Changqi Zhou); Writing—review & editing, Y.L. and C.Z. (Changqi Zhou); Supervision, C.Z. (Changqi Zhou); Project administration, C.Z. (Changqi Zhou), Y.F. and C.Z. (Congwen Zou); Funding acquisition, Y.F., W.J. and C.Z. (Congwen Zou). All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest: The authors declare no conflicts of interest.

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